

Predicting Thermodynamic Properties of Ions in Single Solvents and in Mixed Solvents Using a Modified Born Term within the ePC-SAFT Framework

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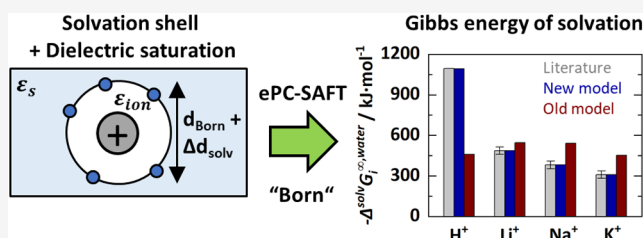
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ABSTRACT: In this work, the EoS ePC-SAFT was further developed by modifying the Born term and introducing a new mixing rule for the dielectric constant of electrolyte solutions. The modified Born term considers the solvent solvation-shell volume of ions and accounts for the radial dependence of the dielectric constant near a central ion. This required one additional parameter per ion and one per solvent, while other pure-component parameters were inherited from the literature. The performance of ePC-SAFT was validated against experimental literature data:

Gibbs energies of solvation and of transfer and MIACs in pure water, methanol, ethanol, and in their mixtures. The new modeling approach improves the accuracy of Gibbs energies at infinite dilution while retaining the ability to model properties of concentrated electrolyte systems. Additionally, ePC-SAFT parameters for V^{3+} and VO^{2+} ions were fitted to density and osmotic-coefficient data of VCl_3 and $VOSO_4$ in water using the new approach.



INTRODUCTION

An increasing interest in environmentally benign technologies has been observed in recent years due to the threat of climate change. This was accompanied by a shift in the energy sector from fossil to renewable energy sources.^{1,2} However, renewable energy sources, such as solar or wind energy, are not able to provide electricity on demand. Even though great progress in energy-storage systems was achieved in the last years, their development and improvement are still ongoing.^{3,4} The most prominent representative of energy-storage systems is the battery, which usually consists of two electrodes and a connecting electrolyte. The composition of these electrolytes ranges from aqueous electrolyte solutions in the case of vanadium-flow batteries to organic electrolyte solutions in Li-ion batteries and even solid-state electrolyte systems.^{5–7} A lot of research interest lies in developing better electrolyte solutions for these systems to reduce viscosity and increase power density, stability, safety, and voltage range of the battery.^{7,8} This includes both the development of better solvents or solvent mixtures and the use of different ions in the solution. One major goal is the replacement of lithium ions in Li-ion batteries with much cheaper and safer materials such as sodium ions.^{3,9} It is of great interest to understand how different ions behave in different solvents for these developments. Electrolyte solutions are also used in many other industrial fields. They are present in (reverse) osmosis processes, carbon-capture technologies, and electrolysis processes.^{10–12} Recent research focuses on replacing chemical processes (fossil-based reactants) with electrochemical

routes.¹³ One example is the production of ammonia using an electrolysis process with a LiOH solution. While originally only LiOH aqueous solution were tested, organic solvents and salt mixtures have also been used in more recent research in this process.^{13–15} Again, knowledge of the interactions between solvents and ions aids the development in these areas. A lot of thermodynamic data (e.g., density, infinite-dilution solvation properties, activity coefficients) on salt-solvent systems can be found in the literature for common electrolytes (e.g., alkali halides) in common solvents (e.g., water, ethanol). However, most literature sources include data on only single-solvent single-electrolyte mixtures. However, data are scarce for an electrolyte in a solvent mixture. Electrolyte thermodynamic models can aid in these cases to describe (or even predict) the properties of more complex mixtures in order to reduce the high experimental effort to characterize multisolvent multielectrolyte mixtures. Many different models exist in the literature for electrolyte solutions.^{16–21} In general, such models were developed to describe aqueous electrolyte solutions, but such models have been subsequently improved to also describe nonaqueous electrolyte solutions. In this context, the implementation of the

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Born term was intensely discussed in the literature, and many models included this term, which resulted in better modeling results in nonaqueous electrolyte solutions.^{16,22–25} The Born term calculates the transfer energy of an ion from a vacuum into any desired solvent, and this energy contribution is usually comparatively high. One of the models that include the Born term is electrolyte PC-SAFT (ePC-SAFT advanced¹⁶), which combines PC-SAFT with Coulomb forces as well as ion-dipolar interactions. It has already been used to model mean ionic activity coefficients (MIACs) and osmotic coefficients in aqueous and organic solvents.^{16,26} Further, ePC-SAFT was used to calculate salt solubilities in these solvents and to predict CO₂ solubilities in aqueous and organic electrolyte solutions^{27,28} and even to predict the salt influence on esterification reaction equilibria and kinetics.^{29,30} All of these works focused on systems with moderate to high ion concentrations. However, the properties of electrolyte solutions at infinite dilution were not investigated in these works. Properties at infinite dilution (like the Gibbs energy of solvation or the Gibbs energy of transfer) are very important, e.g., for the calculation of battery cell voltages in different solvents. The aim of this work is to further improve ePC-SAFT advanced to better describe these infinite-dilution properties of electrolyte solutions while retaining the ability of ePC-SAFT to model properties of concentrated electrolyte solutions.

EPC-SAFT

ePC-SAFT is an extension of the PC-SAFT EoS in order to access the modeling of charged species. In general, PC-SAFT calculates the residual Helmholtz energy a^{res} by evaluating different contributions. These contributions are the hard-chain contribution a^{hc} , the dispersion contribution a^{disp} , and the association contribution a^{assoc} . In ePC-SAFT, two additional contributions for charged species are included: the Debye–Hückel contribution a^{DH} for ion–ion interactions and the Born contribution a^{Born} for ion–dipolar interactions. The residual Helmholtz energy is obtained by summing these contributions (eq 1)

$$a^{\text{res}} = a^{\text{hc}} + a^{\text{disp}} + a^{\text{assoc}} + a^{\text{DH}} + a^{\text{Born}} \quad (1)$$

Original ePC-SAFT requires two ion-specific parameters, while three pure-component parameters for nonassociating components and five pure-component parameters for associating components are required. These are the segment number m_i^{seg} , the segment diameter σ_i , and the dispersion energy parameter u_i/k_B for nonassociating components, where k_B represents the Boltzmann constant. Associating components also require the association-energy parameter ϵ^{AiBi} and the association-volume parameter κ^{AiBi} . Additionally, the binary interaction parameter k_{ij} between every component pair might be adjusted to alter the dispersion energy in the mixture. The corresponding combining rule is given in eq 2

$$u_{ij} = \sqrt{u_i u_j} \cdot (1 - k_{ij}) \quad (2)$$

Note that the binary interaction parameter k_{ij} might depend on the temperature.

Both the expressions for a^{DH} and a^{Born} require a diameter that characterizes the ionic species. In all former ePC-SAFT modeling approaches, the temperature-corrected segment diameter of the ion d_{ion} was used in both expressions, a^{DH} and a^{Born} . eq 3 shows the general formula for temperature-corrected segment diameter $d_i(T)$

$$d_i(T) = \sigma_i \cdot (1 - 0.12 \cdot e^{-3 \cdot u_{ii}/k_B T}) \quad (3)$$

This equation can be simplified for ions since the dispersion energy between cation–cation pairs and anion–anion pairs is set to zero in ePC-SAFT ($u_{ii} = 0$), which leads to eq 4

$$d_{\text{ion}} = \sigma_{\text{ion}} \cdot 0.88 \quad (4)$$

In this work, no changes were made in the expressions for a^{DH} , and d_{ion} is still used in a^{DH} . However, a new diameter parameter for the Born term is introduced in this work. This parameter was not tested at different temperatures and did not include a temperature dependency.

BORN TERM

The Born term is used in many thermodynamic models to calculate the transfer work of an ion from a vacuum into a pure solvent.^{16,23,31} In the Born term, the solvent is treated as a continuum that only interacts with the ion via its dielectric constant ϵ_r .³² The energy connected to this process of a single ion can be summed for all ions in the system, yielding the reduced Helmholtz energy contribution from Born term a^{Born} , given in eq 5

$$a^{\text{Born}} = -\frac{e^2}{4\pi\epsilon_0 k_B T} \left(1 - \frac{1}{\epsilon_r}\right) \sum_i \frac{x_i z_i^2}{d_i^{\text{Born}}} \quad (5)$$

Here, d_i^{Born} is the Born diameter of ion i . This quantity is not well-defined, and many different possibilities exist for its determination. In all former ePC-SAFT modeling approaches, d_i^{Born} was set to σ_i in order to reduce the number of the model parameters. However, this led to deviations between the modeled and experimental Gibbs energies of solvation at infinite dilution of the considered ion, $\Delta^{\text{solv}} G_i^\infty$.¹⁶ Other authors used Pauling diameters or cavity diameters to estimate d_i^{Born} , which also did not yield optimal results for $\Delta^{\text{solv}} G_i^\infty$ when compared to literature data.^{33–35} Some authors suggested fitting d_i^{Born} to literature $\Delta^{\text{solv}} G_i^\infty$ data.^{34,36} Another approach is to not only look at the diameter of the ion but also include the solvent solvation shell in the Born term.^{34,35,37–40} This is done by introducing an additional quantity into Born term Δd_i that accounts for this effect. This quantity might depend on the kind of ion, the solvent, and the composition of the system. Approaches that include this quantity will be called “solvation shell models” (SSMs) in this work. The formula for the reduced Helmholtz energy contribution from the Born term with an SSM approach is given in eq 6

$$a_{\text{SSM}}^{\text{Born}} = -\frac{e^2}{4\pi\epsilon_0 k_B T} \left(1 - \frac{1}{\epsilon_r}\right) \sum_i \frac{x_i z_i^2}{d_i^{\text{Born}} + \Delta d_i} \quad (6)$$

An SSM assumes that the volume between the surface of the ion and the first solvation shell does not contribute to the Helmholtz energy. This assumption does not hold when looking at the changes in the dielectric constant around an ion. The dielectric constant is much lower close to an ion in a solvent environment compared to the bulk phase since the solvent molecules do not affect the electric fields at those distances.^{41–43} This effect is called “dielectric saturation” (DS) and can be included in the Born term by introducing a function for the dielectric constant that depends on the distance from the ion. While it is possible to obtain a radius-dependent function for the dielectric constant, the effort is very high, especially if different solvents, solvent mixtures, and

concentrated electrolyte solutions are considered. An easier equation can be obtained by assuming a constant dielectric constant between the ion surface and the solvation shell that increases instantaneously to the bulk dielectric constant at distances greater than the radius of the solvation shell.^{39,40} In this work, the dielectric constant between the ion and the first solvation shell is assumed to be equal to the dielectric constant of the ion. The value of the relative dielectric constant of all ions is assumed to be $\epsilon_r = 8$ in the ePC-SAFT calculations. This value was determined by Bülow et al. as the average of the relative dielectric constant of different salts taken from the literature.^{16,44} The simplified dependence of the relative dielectric constant on the radius is schematically shown in Figure 1. Both the solvation shell and the dielectric saturation

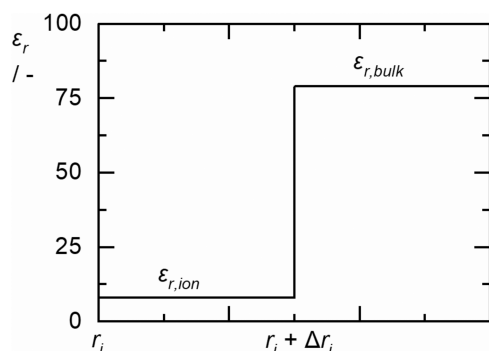


Figure 1. Relative dielectric constant ϵ_r as a function of the radius r as assumed in combined solvation shell and dielectric saturation models. r_i is the radius of the ion, and Δr_i is the distance between the ion surface and the first solvation shell.

effect are included in this combined approach (SSM + DS), which results in eq 7 for the reduced Helmholtz energy contribution of the Born term.

$$a_{\text{SSM+DS}}^{\text{Born}} = -\frac{e^2}{4\pi\epsilon_0 k_B T} \left(1 - \frac{1}{\epsilon_{r,\text{bulk}}} \right) \left(\sum_i \frac{x_i z_i^2}{d_i^{\text{Born}} + \Delta d_i} \right) + \left(1 - \frac{1}{\epsilon_{r,\text{bulk}}} \right) \left(\sum_i x_i z_i^2 \left(\frac{1}{d_i^{\text{Born}}} - \frac{1}{d_i^{\text{Born}} + \Delta d_i} \right) \right) \quad (7)$$

For SSMs, the Born diameter and the solvation shell thickness have the same effect on the Helmholtz energy (eq 6), and it could be argued that the quantities cannot be considered independently. In contrast, for SSM+DS, a separate consideration for the Born diameter d_i^{Born} and the combined diameter $d_i^{\text{Born}} + \Delta d_i$ is necessary since the consideration of the dielectric saturation introduces a contribution that only depends on the Born diameter (eq 7).

It is also possible in these kinds of models to explicitly consider more than one solvation shell layer.⁴⁵ However, it was found that the performance does not significantly improve compared to the models that use only one solvation shell layer.⁴⁶ This consideration might not even be necessary if parameters are fitted to literature data since these data include all solvation shell layers that might exist and therefore can only yield parameters that implicitly include these effects. A comparison of the performance of the three approaches (original, SSM, SSM+DS) is given in Chapters S1 and S2 in

the Supporting Information. The SSM + DS approach was chosen for all further calculations due to its superior performance compared with the other approaches.

In the literature, numerous approaches exist to calculate the solvation shell diameter. However, most of them only consider ions at infinite dilution, and such approaches yield poor results for concentrated solutions or are only applicable to water (for details, see Chapters S1 and S2 in the Supporting Information). Thus, a new parameter f_{mix} was included in ePC-SAFT that depends on the solvent as well as on the ion concentration but not on the kind of ion. The solvation shell thickness was calculated using eq 8

$$\Delta d_i = \frac{(f_{\text{mix}} - 1)}{|z_i|} \cdot d_i^{\text{Born}} \quad (8)$$

The parameter $f_{\text{solv,mix}}$ was calculated via a mixing rule (eq 9)

$$f_{\text{mix}} = \frac{\sum_k x_k f_k}{\sum_k x_k} \quad (9)$$

Here, k is the number of ions and solvents in the system, and f_k is the “pure-component” adjustable parameter for each of these components. The f_k parameter was set to $f_k = 1$ for all ions since no solvation shell exists in a system consisting of pure ions. For now, it was chosen to include only ions and solvents in this mixing rule. This leads to a simplification of eq 9 since no other components are considered in this work (eq 10)

$$f_{\text{mix}} = \sum_k x_k f_k \quad (10)$$

The parameter for each solvent was adjusted to MIACs of NaBr in the respective solvent considered in this work (water, methanol, and ethanol). It should be noted that the f_k value is constant for each solvent. In eq 8, Δd_i is normalized to the absolute charge of the respective ion i . This caused significantly better modeling results for multivalent ions.

DIELECTRIC CONSTANT

An important quantity for modeling electrolyte solutions with ePC-SAFT is the dielectric constant. The relative dielectric constants of all components modeled in this work are given in Table 1 at 298.15 K and 1 bar. All ions were characterized with

Table 1. Relative Dielectric Constants of All Solvents and Ions Used in This Work Are Shown at 298.15 K and 1 bar

component	dielectric constant (–)	refs
water	78.09	47
methanol	33.05	48
ethanol	24.88	49
salts (Ions)	8	16 ^a

^aAll salts were modeled with a similar dielectric constant that is a mean of available experimental data.⁴⁴

the same value of 8 for the relative dielectric constant, which is the average of a number of different salts. This approach was also used in earlier versions of ePC-SAFT.¹⁶

Earlier modeling approaches using the ePC-SAFT framework assumed a linear dependency of the relative dielectric constant on the ion mole fraction.^{16,50,51} This assumption was validated against mixtures of ionic liquids with water. However, a linear dependency of the relative dielectric constant on the ion mole fraction cannot be observed in electrolyte solutions

that include salts, e.g., alkali halides.^{52–60} The relative dielectric constant decreases strongly at low ion mole fractions and gets less steep at higher ion mole fractions in solvents (in this work: water, methanol, and ethanol). A new mixing rule based on the ideas of Zuber et al. is introduced in this work to capture this behavior.⁵² For that purpose, relative dielectric constants in water and methanol were correlated over the ion mole fraction. This new mixing rule was also used to calculate the relative dielectric constants of salt solutions in ethanol to validate it in a different solvent that was not used in the correlation. The new mixing rule is given in eq 11

$$\epsilon_{r,\text{mix}} = \frac{\epsilon_{r,\text{solvent,mix}}^{\text{salt-free}}}{1 + 7.01 \cdot x_{\text{ion}}} \quad (11)$$

Here, $\epsilon_{r,\text{solvent,mix}}^{\text{salt-free}}$ is the relative dielectric constant of the solvent mixture without ions, and x_{ion} is the mole fraction of all ions in the system. In Figure 2, the results of this mixing rule are

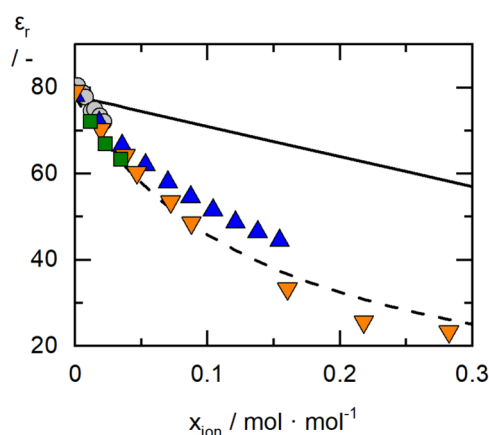


Figure 2. Relative dielectric constants of different aqueous salt solutions at 298.15 K and 1 bar. Solid line: linear mixing rule; dashed line: new mixing rule introduced in this work by eq 11. Gray circles: Data of NaCl in water by Nörtemann et al.⁵³ Green squares: Data of NaBr in water by Haggis et al.⁵⁵ Blue triangles: Data of NaCl in water by Buchner et al.⁵⁶ Orange upside-down triangles: Data of LiCl in water by Wei et al.⁵⁴.

compared to experimental relative dielectric constants of salt solutions in pure water as well as to the linear mole fraction-based mixing rule.

The new mixing allows mimicking the experimental data of numerous salts in water sufficiently well, while the previously used linear mixing rule overestimates the experimental data in the considered mole fraction range. Figure 3 shows the performance of the new mixing for electrolyte solutions in water, methanol, and ethanol. The dielectric constants in all three solvents can be described without using solvent-specific parameters or ion-specific parameters. However, it should be mentioned that this new mixing rule was not tested in any other solvent, and the results might be poor for solvents with very low dielectric constants.⁶¹ Additionally, this new mixing rule was correlated using experimental data points up to about $x_{\text{ion}} = 0.38$ and might deviate from experimental data points at higher ion concentrations. This is acceptable since the solubility of most salts is much lower than this limit. It is expected that the mixing rule is valid for temperatures other than 298.15 K only if the temperature influence on the pure solvents' dielectric constants is considered quantitatively

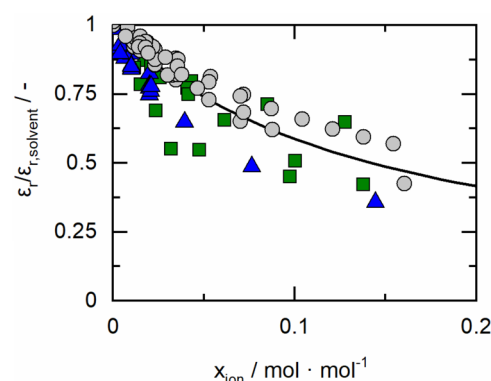


Figure 3. Relative dielectric constants of different salt solutions in different solvents over the relative dielectric constant of the pure solvent at 298.15 K and 1 bar. Solid line: Mixing rule introduced in this work by eq 11; gray circles: Data of different salts in water taken from refs 41,53–56; blue triangles: Data of different salts in methanol taken from refs 57,58; green squares: Data of different salts in ethanol taken from refs 59,60.

correct. However, this hypothesis was not tested in this work since all systems were investigated at 298.15 K.

Similar to earlier ePC-SAFT works, the dielectric constant of a solvent mixture $\epsilon_{r,\text{solvent,mix}}^{\text{salt-free}}$ was obtained by applying a weight fraction-based mixing rule (eq 12) to the pure-component dielectric constants⁵⁰

$$\epsilon_{r,\text{solvent,mix}}^{\text{salt-free}} = \sum_{\text{solvent}} w_{\text{solvent}}^{\text{salt-free}} \cdot \epsilon_{r,\text{solvent}} \quad (12)$$

Here, $w_{\text{solvent}}^{\text{salt-free}}$ is the weight fraction of each solvent in the system without salt. This equation (together with eq 11) allows the estimation of the relative dielectric constant in all multicomponent systems, given that the pure component dielectric constants are known (which in this work were taken from Table 5).

CALCULATION OF THERMODYNAMIC PROPERTIES

The fugacity coefficient of each component in a system can be calculated if the residual Helmholtz energy and its derivations are known by using eq 13

$$\ln(\phi_i) = (Z - 1) - \ln(Z) + a^{\text{res}} + \left(\frac{\partial a^{\text{res}}}{\partial x_i} \right) - \sum_j x_j \left(\frac{\partial a^{\text{res}}}{\partial x_j} \right) \quad (13)$$

The fugacity coefficient can be used to calculate different quantities of the system.

Solvation describes the process of transferring a species from a pure, ideal gaseous phase (g) into a liquid phase (l). The corresponding Gibbs energy of solvation of a species $\Delta^{\text{solv}} G_i(T, p, x_i)$ can be calculated as the difference of its chemical potential μ_i in both phases as given in eq 14.

$$\Delta^{\text{solv}} G_i(T, p, x_i) = \mu_i^{\text{l}}(T^{\text{l}}, p^{\text{l}}, x_i^{\text{l}}) - \mu_{0i}^{\text{id,g}}(T^{\text{g}}, p^{\text{g}}) \quad (14)$$

The temperature is usually the same for both phases, while the pressure is often different for both phases depending on the chosen standard process. The role of the different standard processes is explained later in this chapter. The chemical

potential of a species i can be expressed in the following way (eq 15):

$$\mu_i(T, p, x_i) = \mu_{0i}^{\text{id,g}}(T, p) + RT \ln \left(\frac{f_i}{p} \right) \quad (15)$$

Here, f_i is the fugacity of component i in the mixture, and $\mu_{0i}^{\text{id,g}}(T, p)$ is the chemical potential of a pure, ideal gas at pressure p and temperature T . The fugacity can be expressed via fugacity coefficient φ_i and mole fraction x_i as shown in eq 16

$$f_i = \varphi_i x_i p \quad (16)$$

In theory, the pressure of the gaseous phase and the concentration in the liquid phase can be arbitrary. However, different standard treatments have been established that yield different Gibbs energies. The three most common standard processes found in the literature are the ρ -treatment, the x -treatment, and the m -treatment. The ρ -treatment corresponds to a transfer from an ideal gas at a given concentration (usually 1 mol/L) to a hypothetical solution with the same solute concentration at 1 bar pressure and the properties of an infinitely diluted system.⁶² Note that the pressure in both phases is not equal in this treatment. The x -treatment describes the transfer “from an ideal gas at 1 atm (1 bar in this work) pressure to a hypothetical dilute-ideal solution (at the same pressure) in which the mole fraction (of the solute) s is unity” (see⁶² page 16). The m -treatment corresponds to a transfer from an ideal gaseous phase at 1 bar pressure to a hypothetical solution at the same pressure with a solute concentration of 1 mol/kg and the properties of an infinitely diluted system.⁶² Note that this concentration is given in the molality scale and refers to 1 mol solute per 1 kg pure solvent (or solvent mixture). All three treatments assume the same temperature in the gaseous and liquid phases. In this work, the Gibbs energies of solvation are given for the x -treatment, and literature data was converted to this treatment when necessary. The Gibbs energy of solvation for the x -process was obtained by combining eqs 14–16 and is given in eq 17

$$\begin{aligned} \Delta^{\text{solv},x} G_i^{\infty}(T, p, x_{j \neq i}, x_i \rightarrow 0) &= \mu_{0i}^{\text{id,g}}(T, p) \\ &+ RT \ln \left(\frac{\varphi_i^{\infty} x_i p}{p} \right) - \mu_{0i}^{\text{id,g}}(T, p) \\ &= RT \ln (\varphi_i^{\infty}(T, p, x_{j \neq i}, x_i \rightarrow 0)) \end{aligned} \quad (17)$$

Here, φ_i^{∞} is the fugacity coefficient of the ion at infinite dilution, which corresponds to the properties at infinite dilution as mentioned in the definition of the x -treatment. Therefore, the superscript ∞ is added to the Gibbs energies in this work. To convert values of the Gibbs energy of solvation from the ρ -treatment or the m -treatment, eq 18 can be used.

$$\begin{aligned} \Delta^{\text{solv},x} G_i^{\infty}(T, p, x_{j \neq i}, x_i \rightarrow 0) &= \Delta^{\text{solv},\rho} G_i^{\infty}(T, p, x_{j \neq i}, x_i \rightarrow 0) \\ &+ RT \ln \left(\frac{RT \rho_{\text{solvent}}}{p^0 M_{\text{solvent}}} \right) \\ &= \Delta^{\text{solv},m} G_i^{\infty}(T, p, x_{j \neq i}, x_i \rightarrow 0) \\ &+ RT \ln \left(\frac{1}{\tilde{m}^0 M_{\text{solvent}}} \right) \end{aligned} \quad (18)$$

Here, p^0 is the chosen standard pressure for the x -process (usually 1 bar), $\tilde{m}^0$ is the standard molality (1 mol/kg), ρ_{solvent} the density of the solute-free solvent (or solvent mixture), and M_{solvent} is the molar mass of the solvent (or solvent mixture). A detailed deviation of the conversion is found in the Supporting Information in Chapter S3. The superscript x for the x -process will be omitted in the following text since only values for the x -process are presented.

The Gibbs energy of transfer at infinite dilution from a solvent S1 to another solvent or solvent mixture S2 $\Delta^{\text{tr}} G_i^{\infty, S1 \rightarrow S2}$ was calculated as the difference of the Gibbs energies of solvation in the two solvents (eq 19)

$$\begin{aligned} \Delta^{\text{tr}} G_i^{\infty, S1 \rightarrow S2}(T, p, x_{j \neq i}, x_i \rightarrow 0) &= \Delta^{\text{solv}} G_i^{\infty, S2}(T, p, x_{j \neq i}, x_i \rightarrow 0) \\ &- \Delta^{\text{solv}} G_i^{\infty, S1}(T, p, x_{j \neq i}, x_i \rightarrow 0) \\ &= RT \ln \left(\frac{\varphi_i^{\infty, S2}}{\varphi_i^{\infty, S1}} \right) \end{aligned} \quad (19)$$

The activity coefficients of ions were calculated by using eq 20

$$\gamma_i^* = \frac{\varphi_i^S}{\varphi_i^{\infty, S}} \quad (20)$$

The “*” indicates that the reference state for the activity coefficient is the infinitely diluted solution, which is most commonly used for ions. The mean ionic activity coefficient (MIAC) $\gamma_{\pm}^*$ is usually given in the literature. It can be calculated from the single ion activity coefficients using eq 21

$$\gamma_{\pm}^* = (\gamma_{\text{cation}}^* \gamma_{\text{anion}}^*)^{1/(\nu_{\text{cat}} + \nu_{\text{an}})} \quad (21)$$

Here, ν_{cat} and ν_{an} are the stoichiometric coefficients of the ions. MIACs are usually given in the molality scale $\gamma_{\pm}^{*,m}$ in the literature while ePC-SAFT calculates mole fraction-based MIACs $\gamma_{\pm}^{*,x}$. They were converted using eq 22

$$\gamma_{\pm}^{*,m} = \frac{\gamma_{\pm}^{*,x}}{1 + M_{\text{solvent}} \cdot \tilde{m}_{\text{solute}} \cdot \sum_i \nu_{i, \text{ion}}} \quad (22)$$

Here, M_{solvent} is the molar mass of the solvent (in SI units), and $\tilde{m}_{\text{solute}}$ is the molality of the nondissociated solute. In this work, all MIACs are given in the molality scale.

MODEL PARAMETERS

The pure-component parameters for the ions and solvents were taken from the literature, with the exception of the parameters for V^{3+} and VO^{2+} . These were fitted in this work to our own density data of VCl_3 and VOSO_4 in water (see

Chapter S4 in the Supporting Information for additional details). The new modeling approach used in this work requires two additional parameters: The Born diameter d_i^{Born} of ions and the solvation parameter f_k for solvents (set to 1 for all ions). The Born diameters were adjusted to Gibbs energies of solvation at infinite dilution in water (details on the used Gibbs energies of solvation in water are found in Chapter S5 in the Supporting Information), and the solvent-specific (ion-independent) f_k parameters were adjusted to MIAC data of NaBr in each respective solvent. It can be observed that f_k is close to 1.5 for the considered solvents. Indeed, we found that $f_k = 1.5$ for the considered solvents if only electrolyte solutions with low ion concentrations are considered (ca. 10 mM). The influence of f_k on $\Delta^{\text{solv},x}G_i^\infty$ is stronger with the smaller d_i^{Born} caused by the dependency of a^{Born} on $1/d_i^{\text{Born}}$, cf. eq 7. Thus, the influence of f_k on MIACs is much more pronounced than that on $\Delta^{\text{solv},x}G_i^\infty$.

New binary interaction parameters k_{ij} between the ions and the solvents and between all ions were also adjusted in this work. The k_{ij} between an ion and water and the k_{ij} among ions were adjusted to experimental MIAC literature data of different salts in water. In the special case of V^{3+} and VO^{2+} , the k_{ij} parameters V^{3+} -water, VO^{2+} -water, V^{3+} - Cl^- , and VO^{2+} - SO_4^{2-} were fitted to new osmotic coefficients of VCl_3 and VOSO_4 in water.

This order for fitting the different parameters (d_i^{Born} , f_k , and k_{ij} between ions and water) was chosen for the following reasons: The Gibbs energy of solvation of an ion in water is strongly influenced by the d_i^{Born} parameter and only slightly changes with changing f_k and k_{ij} . The d_i^{Born} parameter was therefore fit to this quantity, while MIACs in water were used to adjust the other parameters. An iteration was then performed to get an optimal fit for all quantities. However, the change in all parameters during the iteration was minimal.

Gibbs energies of solvation at infinite dilution in organic solvents were calculated using the literature Gibbs energies of solvation in water and the literature Gibbs energies of transfer from water to the solvent. These data in the pure organic solvents were used to determine the k_{ij} between ion/methanol and ion/ethanol. k_{ij} values for solvent–solvent pairs were taken from the literature since the new modeling approach does not affect systems that only consist of neutral components. All pure-component parameters are given in Tables 2 and 3, and all binary interaction parameters used in this work are presented in Tables 4 and 5. The values of the k_{ij} parameters between the ions and the solvents deviate strongly from 0, especially between an ion and an organic solvent. This strong deviation can be explained by the low sensitivity of the dispersion interactions compared to ion-dipolar forces

Table 2. Pure-Component Parameters of All Solvents Used in This Work^a

organic solvent	water ⁶³	methanol ⁶⁴	ethanol ⁶⁴
m_i^{seg}	1.2047	1.5255	2.3827
$\sigma_i/\text{\AA}$	^b	3.2300	3.1771
$u_i/k_B/\text{K}$	353.95	188.90	198.24
$\epsilon^{\text{AiBi}}/\text{K}$	2425.7	2899.5	2653.4
κ^{AiBi}	0.04509	0.03518	0.03238
f_k	1.5	1.4	1.6

^aAll solvents have a 2B association scheme. ^b $\sigma = 2.7927 + (10.11 \cdot e^{-0.01775 \text{ T/K}} - 1.417 \cdot e^{-0.01146 \text{ T/K}})$

Table 3. ePC-SAFT Pure-Ion Parameters, σ_i and u_i/k_B , Taken from^{a17}

ion	$\sigma_i/\text{\AA}$	$u_i/k_B/\text{K}$	$d_i^{\text{Born}} (\text{\AA})$
H^+	3.4654	500.00	1.218
Li^+	2.8449	360.00	2.784
Na^+	2.8232	230.00	3.445
K^+	3.3417	200.00	4.150
VO^{2+}	3.4638	1488.28	2.930
V^{3+}	3.1401	1415.66	2.940
Cl^-	2.7560	170.00	4.100
Br^-	3.0707	190.00	4.480
I^-	3.6672	200.00	4.985
SO_4^{2-}	2.6491	80.00	4.953

^aSegment number is 1 for all ions.

Table 4. ePC-SAFT Binary Interaction Parameters k_{ij} between Different Anions and Cations Fitted in This Work Are Valid at 298.15 K

$k_{\text{cat},\text{an}}$	H^+	Li^+	Na^+	K^+	VO^{2+}	V^{3+}
Cl^-	−0.9	0.8	0.8	0		1
Br^-	−0.7	0.5	0.65	−0.35		
I^-			0.45	0		
SO_4^{2-}		0.8	0.7	−0.6	0	

Table 5. ePC-SAFT Binary Interaction Parameters k_{ij} between Different Components at 298.15 K Used in This Work^a

k_{ij}	water	methanol	ethanol
Cl^-	−0.3	0.5	0.8
Br^-	−0.3	0.15	0
I^-	−0.05	0.37	0.18
SO_4^{2-}	−0.1		
H^+	0	−0.3	−0.6
Li^+	−0.4	−0.9	−0.8
Na^+	−0.3	−0.25	0.05
K^+	−0.1	0.32	0.53
VO^{2+}	−0.08		
V^{3+}	−0.3		
methanol	−0.0878 ⁶⁵		
ethanol	−0.0617 ⁶⁶		

^aAll k_{ij} s between ions and solvents were fitted in this work, and the k_{ij} s between different solvents were inherited from the literature. Valid at 298.15 K

described by the Born term, which was also found in ref 16. Therefore, high k_{ij} are needed to have an effect on the modeling results.

■ GIBBS ENERGY OF SOLVATION IN WATER

The Gibbs energy of solvation at infinite dilution in water $\Delta^{\text{solv}}G_i^{\infty,\text{water}}$ was calculated by using the parameters from Tables 2 and 1. It is clear that the different mixing rules for the dielectric constant have no influence on the Gibbs energies of solvation since that is a property at infinite dilution. The results of $\Delta^{\text{solv}}G_i^{\infty,\text{water}}$ using the earlier version ePC-SAFT advanced and the new ePC-SAFT modeling approach are shown in Figure 4 together with literature data. The Gibbs energy of solvation at infinite dilution in water becomes less negative from Li^+ to K^+ and also from Cl^- to I^- . Larger ions have a lower charge density at the surface, which leads to weaker

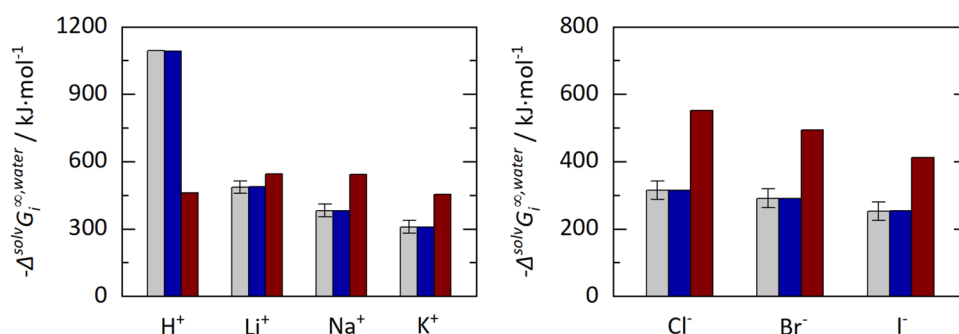


Figure 4. Gibbs energy of solvation at infinite dilution in water of different ions at 298.15 K and 1 bar. Gray: Average value of different literature sources^{36,38,40,67–69} for H^+ only the value of Tissandier et al.⁶⁹ is shown as recommended by Klinksieck et al.;⁷⁰ blue: New modeling approach introduced in this work using parameters from Tables 1–5; and brown: ePC-SAFT advanced results taken from ref 16. All values are given in the mole fraction scale.

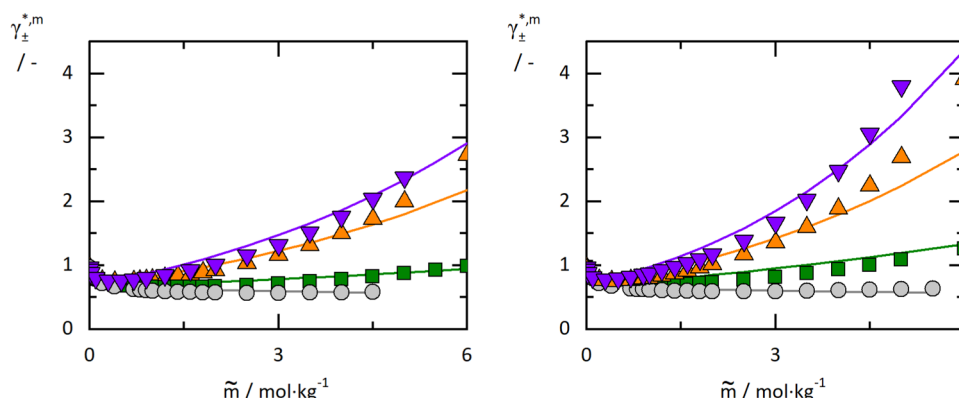


Figure 5. Molality-based MIACs of different salts in water at 298.15 K and 1 bar. Left: Cl^- salts; right: Br^- salts. Symbols: Literature data taken from.⁷¹ Purple upside-down triangles: H^+ components; orange triangles: Li^+ salts; green squares: Na^+ salts; and gray circles: K^+ salts. Lines: Calculated using ePC-SAFT with the modeling approach introduced in this work using parameters from Tables 1–5.

interactions and consequently to Gibbs energies of solvation closer to zero.

The calculations with the new modeling approach are in good agreement with the literature values for $\Delta^{\text{solv}}G_i^{\infty, \text{water}}$. The reason behind this is that $\Delta^{\text{solv},x}G_i^{\infty}$ is sensitive to the Born diameter. ePC-SAFT advanced predicts reasonable $\Delta^{\text{solv}}G_i^{\infty, \text{water}}$ values for Li^+ but is not able to accurately describe this quantity for other ions. This is related to the use of segment diameters in the Born term in ePC-SAFT advanced since those differ greatly from the fitted Born diameters (with the exception of Li^+). Rashin et al. and Schmid et al. also used fitted Born diameters in the same range as the fitted ones in this work.^{34,36} The differences in the exact values of the Born diameters can be attributed to the used literature Gibbs energies of solvation, and the fact that in this work, other contributions than the Born term are included in calculating this quantity. Fawcett et al. calculated the Gibbs energies of solvation of different ions in water and used Shannon-Prewitt crystal radii in the Born term, which are smaller than the fitted values in this work and the works of Rashin et al. and Schmid et al.³⁸ This led to Gibbs energies of solvation that were much more negative than the literature values. Other authors used covalent radii or ionic (Pauling) radii, which also resulted in too negative Gibbs energies of solvation.^{33,34} Therefore, fitting Born diameters to Gibbs energies of solvation is justified, especially since the values in this work are comparable to the values of other authors. However, this again shows the poor definition of the Born diameter and raises questions on its

physical interpretation. The modeled Gibbs energies of solvation at infinite dilution in water for all investigated ions are found in the Supporting Information in Table S4.

MIACS IN WATER

Most applications that use electrolyte solutions operate at higher electrolyte concentrations and cannot be described using only infinite-dilution properties. ePC-SAFT advanced showed great results in modeling concentrated aqueous salt solutions and represents a good benchmark for the new modeling approach. It should again be noted that the aim of this work is the description of properties at infinite dilution while retaining the ability of ePC-SAFT to describe concentrated electrolyte solutions quantitatively correct. Figure 5 shows the MIACs of different salts in water. The new modeling approach is able to describe the initial decrease and the later increase of the MIACs with higher salt concentrations very well for the investigated salts. However, a re-evaluation of the binary interaction parameters compared to earlier versions of ePC-SAFT was necessary to achieve the quantitative agreement. In contrast, the new treatment did not require refitting the pure-ion parameters. Note, the f_k parameter for water was only fitted to MIAC data of NaBr in water, and it remains constant for all other calculations.

PROPERTIES IN ORGANIC AND MIXED SOLVENTS

The next step in this work was the evaluation of infinite-dilution properties of electrolytes in organic solvents. Methanol

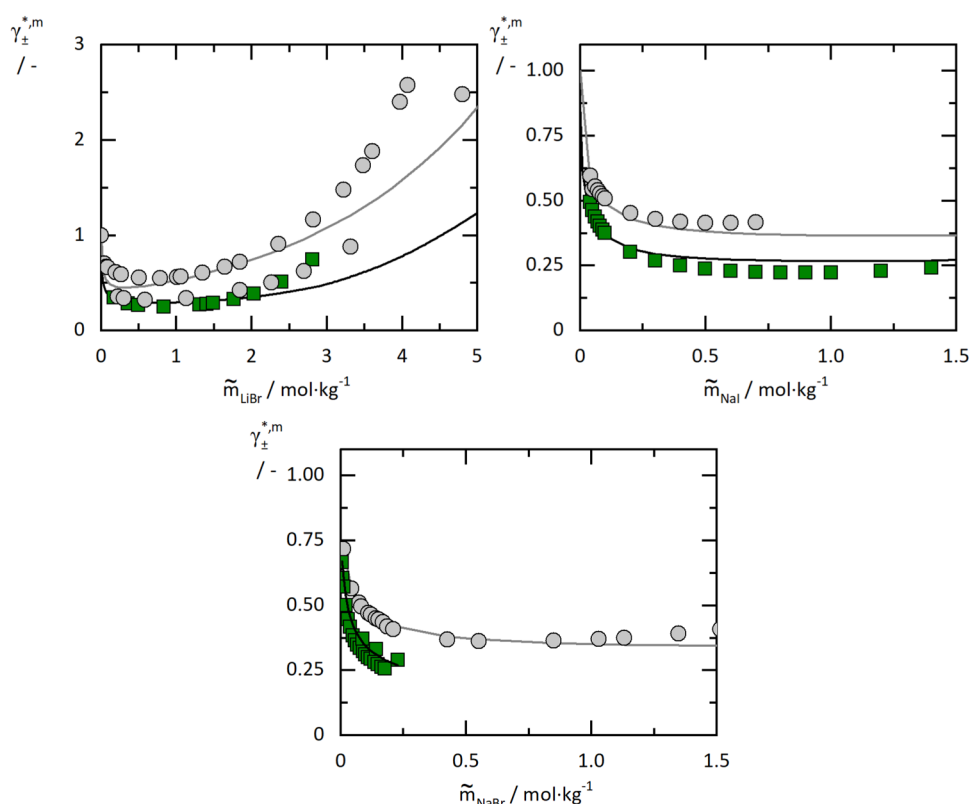


Figure 8. Molality-based MIACs of LiBr (top left), NaI (top right), and NaBr (bottom) in methanol and in ethanol at 298.15 K and 1 bar. Gray circles: Literature MIAC data in methanol;^{75–78} green squares: Literature MIAC data in ethanol;^{77,79–81} lines: ePC-SAFT with the modeling approach introduced in this work using parameters from Tables 1–5; gray line: Methanol; and black line: Ethanol. The MIACs were modeled up to the solubility limit of the salts in the respective solvent as measured in refs 82,83.

fraction-based linear mixing rule for the dielectric constant yielded high deviations between ePC-SAFT and the experimental MIACs.

As a next step, the MIACs of all other salts in ethanol and methanol were modeled without fitting further parameters since the binary interaction parameters ion-methanol and ion-ethanol were fitted to Gibbs energies of solvation at infinite dilution. The MIACs of LiBr, NaI, and NaBr in pure methanol and pure ethanol are shown in Figure 8. Two different sources for LiBr MIACs in methanol were found in the literature.^{75,76} The ePC-SAFT-predicted MIACs agree well with these two data sources up to a molality of about 3.5 mol kg^{−1}. At molalities greater than 3.5 mol kg^{−1}, ePC-SAFT underpredicts the experimental MIACs, which is quite obvious for LiBr in ethanol. These deviations might be explained by the fact that the binary interaction parameters ion-methanol and ion-ethanol were fitted to infinite-dilution data. Note that electrolyte solubility in organic solvents is usually not very high, so a low-to-moderate concentration range is more important.

Further, the ability of ePC-SAFT to predict MIACs in mixed solvent systems was evaluated. Figure 9 shows selected MIACs in water–methanol and water–ethanol mixtures at different solvent compositions. Again, the new modeling approach yields good results for the investigated systems, especially at lower electrolyte concentrations. The modeling results for the system water–ethanol–NaBr show deviations at ion molalities greater than about 2 mol kg^{−1} at salt-free weight fractions of ethanol equal to 0.4. ePC-SAFT underestimates the increase in the MIACs with electrolyte molality at molalities greater than

about 1 mol kg^{−1}. This could be related to the fact that only data at low electrolyte concentrations were used in fitting the parameters for the NaBr–ethanol system (infinite-dilution properties were used for k_{ij} and MIAC data were used up to 0.25 mol kg^{−1} for f_{ethanol}).

To sum up, the new modeling approach within the ePC-SAFT framework allows accurate modeling of properties at infinite dilution and in concentrated electrolyte single-solvent systems and even prediction in mixed-solvent electrolyte systems. The pure-ion parameters could be inherited from earlier works, but the binary interaction parameters needed to be readjusted. The improved performance of ePC-SAFT using our new approach vs recent modeling approaches for infinitely diluted systems was found to be in the new treatment of Born diameters. The good performance of the model for concentrated systems can be attributed to the use of a new mixing rule for the dielectric constant in combination with this new treatment of Born diameter.

CONCLUSIONS

The goal of this work was to increase the accuracy of ePC-SAFT toward modeling properties at infinite dilution in single solvents and in mixed solvents at 298.15 K while keeping the well-known accuracy of describing properties at more concentrated electrolyte solutions. To achieve this goal, ePC-SAFT was further improved by accounting for the solvation-shell thickness and the dielectric saturation effect within the Born term. These modifications required the adjustment of the Born diameter of each ion and one ion-independent solvation parameter per solvent as well as one binary interaction

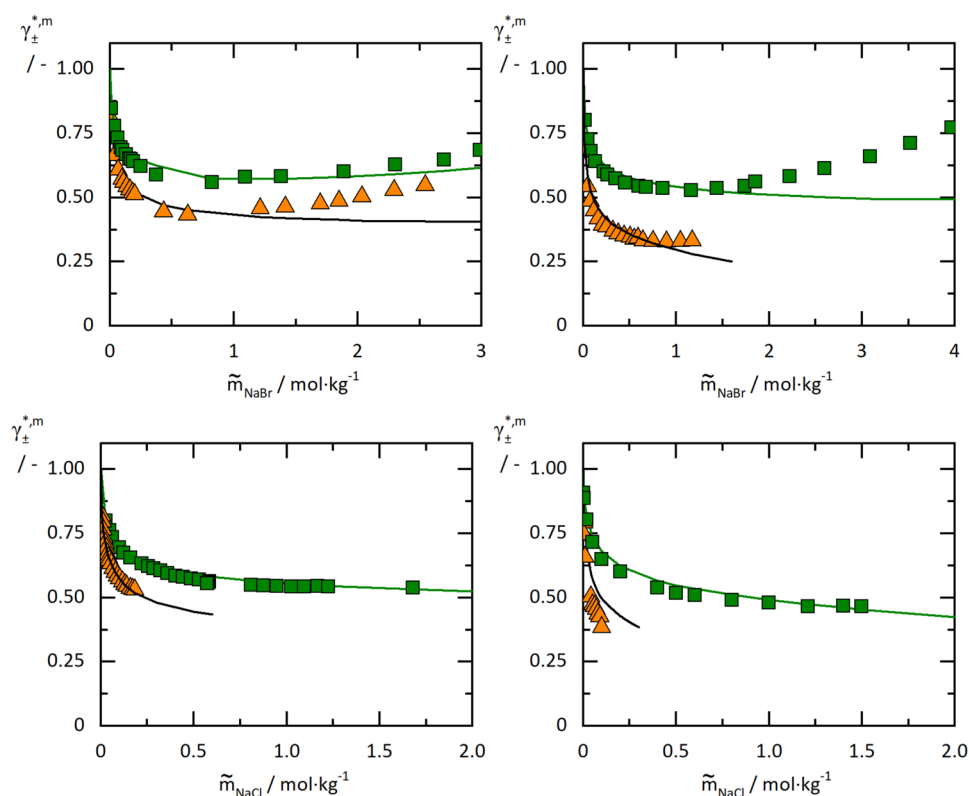


Figure 9. Molality-based MIACs of NaBr (top) and NaCl (bottom) in methanol–water (left) and ethanol–water (right) mixtures at 298.15 K and 1 bar. Orange triangles: Literature MIACs in $w_{\text{organic solvent}}^{\text{salt-free}} = 0.8$ taken from refs 77,84,85; green squares: Literature MIACs in $w_{\text{organic solvent}}^{\text{salt-free}} = 0.4$ taken from refs 77,84,85. Lines are modeling results with ePC-SAFT from this work using parameters from Tables 1–5. The MIACs were modeled up to the solubility of the salts in the respective solvent mixture as measured in refs 82,83.

parameter ion–solvent. It was shown that infinite-dilution properties can be used to adjust the Born diameter, while binary interaction parameters were used to correlate finite-concentration properties (MIACs in this work). All other pure-ion parameters were taken from earlier versions of ePC-SAFT without change. New parameters were determined for vanadium ions V^{3+} and VO^{2+} by fitting to density data and to osmotic-coefficient data measured in this work. Additionally, a new mixing rule for the dielectric constant was introduced by correlating experimental dielectric constant data of single electrolytes in water and in methanol. The mixing rule was then validated and holds true also for single electrolytes in ethanol, which was not used for determining the correlation. The new treatment of the Born term allowed for a better description of properties of ions at infinite dilution in pure and mixed solvents, which was shown by comparing modeled Gibbs energies of solvation and transfer at infinite dilution with literature data. The new modeling approach, including the new mixing rule for the dielectric constant, retained the ability of earlier versions to describe properties of concentrated electrolyte solutions in pure solvents, which was shown on MIACs in water, methanol, and ethanol. Even more, MIACs in mixed solvents were predicted accurately up to molalities of about 2 mol kg^{-1} . This is a significant advance over previous modeling works and should also be helpful for other electrolyte thermodynamic models.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.iecr.5c00475>.

Details on different approaches to calculate the solvation shell diameter; comparison of the performance of different Born contributions in ePC-SAFT together with different mixing rules for the dielectric constant; equations for the conversion of the Gibbs energy of solvation for different standard treatments; density and osmotic coefficient data for aqueous VCl_3 and VO_2SO_4 solutions; comparison of literature data of the Gibbs energy of solvation of different ions in water (PDF)

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Notes

The authors declare no competing financial interest.

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